



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: General Physics B (II) Department: PHYSICS

Exam Duration: 2 hours Exam Paper Setter: Physics teaching team

Question No.	1	2	3	4	5	6	7	8	9	10
Score										

This exam paper contains 8 questions and the score is 100 in total. (Hand in your exam paper, answer sheet, and draft paper to the proctor when the exam ends.)

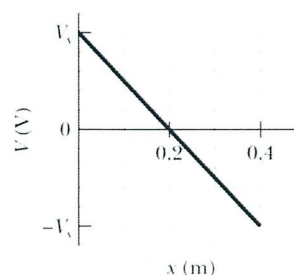
Q1. Multiple Choice Questions ($3 \times 10 = 30$ scores), each question has one correct answer.

1. The electric field due to a uniform distribution of charge on a spherical shell is zero:

- A) everywhere
- B) nowhere
- C) only at the center of the shell
- D) only inside the shell
- E) only outside the shell

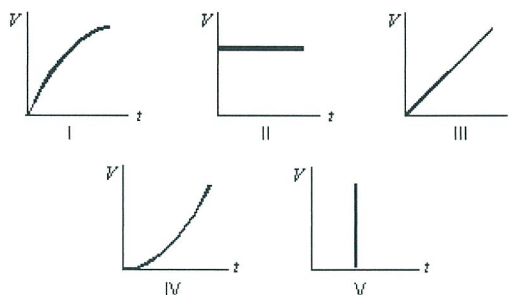
2. The graph shows the electric potential as a function of x in a certain region. What is the x component of the electric field in this region if $V_s = 50$ V?

- A) 250 V/m
- B) 40 V/m
- C) 10 V/m
- D) -40 V/m
- E) -250 V/m



3. Suppose the current charging a capacitor is kept constant. Which graph below correctly gives the potential difference V across the capacitor as a function of time?

- A) I
- B) II
- C) III
- D) IV
- E) V



4. Two long straight wires enter a room through a window. One carries a current of 3.0 A into the room while the other carries a current of 5.0 A out. The magnitude of the path integral $\oint \vec{B} \cdot d\vec{s}$ around the window frame is:

- A) $2.5 \times 10^{-6} \text{ T}\cdot\text{m}$
- B) $3.8 \times 10^{-6} \text{ T}\cdot\text{m}$
- C) $6.3 \times 10^{-6} \text{ T}\cdot\text{m}$
- D) $1.0 \times 10^{-5} \text{ T}\cdot\text{m}$
- E) none of these

5. As a loop of wire with a resistance of $10 \text{ } \Omega$ moves in a non-uniform magnetic field, it loses kinetic energy at a uniform rate of 5 mJ/s. The induced emf in the loop is:

- A) 0 V
- B) 0.22 V
- C) 0.28 V
- D) 2.0 V
- E) cannot be calculated from the given data

6. In a sinusoidally driven series *RLC* circuit, the inductive resistance is $X_L = 200 \text{ } \Omega$, the capacitive reactance is $X_C = 100 \text{ } \Omega$, and the resistance is $R = 50 \text{ } \Omega$. The current and applied emf would be in phase if:

- A) the resistance is increased to $100 \text{ } \Omega$, with no other changes
- B) the resistance is increased to $200 \text{ } \Omega$, with no other changes
- C) the inductance is reduced to zero, with no other changes
- D) the capacitance is doubled, with no other changes
- E) the capacitance is halved, with no other changes

7. Light with an intensity of 1 kW/m^2 falls normally on a surface and is completely reflected. The radiation pressure is:

- A) 1 kPa
- B) $3 \times 10^{11} \text{ Pa}$
- C) $1.7 \times 10^{-6} \text{ Pa}$
- D) $3.3 \times 10^{-6} \text{ Pa}$
- E) $6.7 \times 10^{-6} \text{ Pa}$

8. In a Young's double-slit experiment, a thin sheet of mica is placed over one of the two slits. As a result, the center of the fringe pattern (on the screen) shifts so that the center is now occupied by what was the 30th dark band. The wavelength of the light in this experiment is 480 nm and the index of the mica is 1.60. The mica thickness is:

- A) 0.009 mm
- B) 0.012 mm
- C) 0.014 mm
- D) 0.024 mm
- E) 0.062 mm

9. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygens wavelet from the top of the slit and the wavelet from the midpoint of the slit is:

- A) $\pi/8$ rad
- B) $\pi/4$ rad
- C) $\pi/2$ rad
- D) π rad
- E) $3\pi/2$ rad

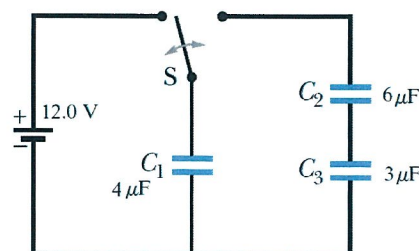
10. A meson when at rest decays $2 \mu\text{s}$ after it is created. If moving in the laboratory at $0.99c$, its lifetime according to laboratory clocks would be:

- A) the same
- B) $0.28 \mu\text{s}$
- C) $4.6 \mu\text{s}$
- D) $14 \mu\text{s}$
- E) none of these

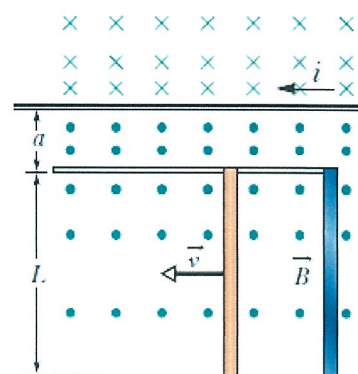
Essay Questions

Q2. (6 scores) Find the maximum kinetic energy of electrons ejected from a certain material if the material's work function is 2.3 eV and the frequency of the incident radiation is $2.5 \times 10^{15} \text{ Hz}$.

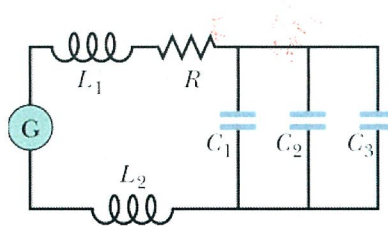
Q3. (10 scores) As shown in the figure, switch is thrown to the left side until C_1 is fully charged. Then the switch is thrown to the right. What is the final charge on C_1 , C_2 and C_3 ?



Q4. (12 scores) The figure shows a rod of length $L = 10.0 \text{ cm}$ that is forced to move at constant speed $v = 5.00 \text{ m/s}$ along horizontal rails. The rod, rails, and connecting strip at the right form a conducting loop. The rod has resistance 0.400Ω ; the rest of the loop has negligible resistance. A current $i = 100 \text{ A}$ through the long straight wire at distance $a = 10.0 \text{ mm}$ from the loop sets up a (nonuniform) magnetic field through the loop. Find: (a) the emf and current induced in the loop. (b) At what rate is thermal energy generated in the rod? (c) What is the magnitude of the force that must be applied to the rod to make it move at constant speed? (d) At what rate does this force do work on the rod?



Q5. (12 scores) In the figure, a generator with an adjustable frequency of oscillation is connected to resistance $R = 100\ \Omega$, inductances $L_1 = 9.70\ \text{mH}$ and $L_2 = 2.30\ \text{mH}$, and capacitances $C_1 = 8.40\ \text{mF}$, $C_2 = 2.50\ \text{mF}$, and $C_3 = 3.50\ \text{mF}$. (a) What is the resonant frequency of the circuit? What happens to the resonant frequency if (b) R is increased, (c) L_1 is increased, (d) C_3 is removed from the circuit, and (e) L_2 is removed?



Q6. (10 scores) In a double-slit arrangement the slits are separated by a distance equal to 100 times the wavelength of the light passing through the slits. (a) What is the angular separation in radians between the central maximum and an adjacent maximum? (b) What is the distance between these maxima on a screen 90.0 cm from the slits?

Q7. (12 scores) How many bright fringes appear between the first diffraction envelope minima to either side of the central maximum in a double-slit pattern if $\lambda = 550\ \text{nm}$, $d = 0.180\ \text{mm}$, and $a = 30.0\ \text{mm}$? What is the ratio of the intensity of the third bright fringe to the intensity of the central fringe?

Q8. (8 scores) An armada (a large group of war ships) of spaceships that is 1.00 ly long (in its rest frame) moves with speed $0.850c$ relative to a ground station in frame S . A messenger travels from the rear of the armada to the front with a speed of $0.950c$ relative to S . How long does the trip take as measured

- (a) in the armada's rest frame
- (b) in the messenger's rest frame,

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Q1. Multiple Choice Questions ($3 \times 10 = 30$ scores)**1-5 DACAB****6-10 EEDDD****Essay Questions****Q2****Solution**

$$K_{\max} = hf - \Phi$$

$$= (4.14 \times 10^{-15} \text{ eV} \cdot \text{s}) (2.5 \times 10^{15} \text{ Hz}) - 2.3 \text{ eV}$$

$$= 8.1 \text{ eV}$$

Q3.**Solution:**

(a) Solving the two equations

$$\frac{q_1}{C_1} = \frac{q_2}{C_{\text{eq}}}, \quad q_1 + q_2 = C_1 V_0$$

for q_1 and q_2 , we obtain

$$q_1 = \frac{C_1^2 V_0}{C_{\text{eq}} + C_1} = \frac{C_1^2 V_0}{\frac{C_2 C_3}{C_2 + C_3} + C_1} = \frac{C_1^2 (C_2 + C_3) V_0}{C_1 C_2 + C_1 C_3 + C_2 C_3}.$$

$$C_{\text{eq}} = 2.0 \mu\text{F} \quad q_1 = 42.7 \mu\text{C} \quad \dots 4'$$

(b) The charge on capacitors 2 is

$$q_2 = C_1 V_0 - q_1 = (4.00 \mu\text{F})(16.0 \text{ V}) - 42.7 \mu\text{C} = 21.3 \mu\text{C} \quad \dots 3'$$

(c) The charge on capacitor 3 is the same as that on capacitor 2:

$$q_3 = C_1 V_0 - q_1 = (4.00 \mu\text{F})(16.0 \text{ V}) - 42.7 \mu\text{C} = 21.3 \mu\text{C} \quad \dots 3'$$

Q4.**Solution:**

- (a) We consider an infinitesimal horizontal strip of length x and width dr , parallel to the wire and a distance r from it; it has area $A = x dr$ and the flux is

$$d\Phi_B = B dA = \frac{\mu_0 i}{2\pi r} x dr$$

$$\Phi_B = \frac{\mu_0 i x}{2\pi} \int_a^{a+L} \frac{dr}{r} = \frac{\mu_0 i x}{2\pi} \ln\left(\frac{a+L}{a}\right).$$

According to Faraday's law the emf induced in the loop is

$$\mathcal{E} = \frac{d\Phi_B}{dt} = \frac{\mu_0 i}{2\pi} = 2.4 \times 10^{-4} \text{ V}$$

By Ohm's law, the induced current is

$$i_t = \mathcal{E} / R = (2.40 \times 10^{-4} \text{ V}) / (0.400 \Omega) = 6.00 \times 10^{-4} \text{ A}.$$

...4'

Since the flux is increasing, the magnetic field produced by the induced current must be into the page in the region enclosed by the rod and rails. This means the current is clockwise.

- (b) Thermal energy is being generated at the rate

$$P = i_t^2 R = (6.00 \times 10^{-4} \text{ A})^2 (0.400 \Omega) = 1.44 \times 10^{-7} \text{ W}.$$

...4'

- (c) The force of the external agent must have the same magnitude as the magnetic force and must be in the opposite direction.

$$dF_B = i_t B dr = (\mu_0 i_t i / 2\pi r) dr.$$

$$\begin{aligned} F_B &= \frac{\mu_0 i_t i}{2\pi} \int_a^{a+L} \frac{dr}{r} = \frac{\mu_0 i_t i}{2\pi} \ln\left(\frac{a+L}{a}\right) \\ &= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6.00 \times 10^{-4} \text{ A})(100 \text{ A})}{2\pi} \ln\left(\frac{1.00 \text{ cm} + 10.0 \text{ cm}}{1.00 \text{ cm}}\right) \\ &= 2.87 \times 10^{-8} \text{ N.} \quad \text{toward the right} \end{aligned}$$

...4'

The external agent must therefore apply a force of $2.87 \times 10^{-8} \text{ N}$, to the left.

- (d) The external agent does work at the rate

$$P = F v = (2.87 \times 10^{-8} \text{ N})(5.00 \text{ m/s}) = 1.44 \times 10^{-7} \text{ W}.$$

...4'

Q5.**Solution:**

(a) Since $L_{eq} = L_1 + L_2$ and $C_{eq} = C_1 + C_2 + C_3$ for the circuit, the resonant frequency is

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(L_1 + L_2)(C_1 + C_2 + C_3)}}$$

$$= \frac{1}{2\pi\sqrt{(9.70 \times 10^{-3} H + 2.30 \times 10^{-3} H)(8.40 \times 10^{-6} F + 2.50 \times 10^{-6} F + 3.50 \times 10^{-6} F)}}$$

$$= 383 \text{ Hz}$$

...4'

(b) The resonant frequency does not depend on R so it will not change as R increases.

...2'

(c) Since $\omega \propto (L_1 + L_2)^{-1/2}$, it will decrease as L_1 increases.

...2'

(d) Since $\omega \propto C_{eq}^{-1/2}$ and C_{eq} decreases as C_3 is removed, ω will increase.

...2'

(e) Since $\omega \propto (L_1 + L_2)^{-1/2}$, it will increase as L_2 is removed.

...2'

Q6.**Solution:**

(a) For the maximum adjacent to the central one, we set $m = 1$ in Eq. 35-14 and obtain

$$\theta_1 = \sin^{-1}\left(\frac{m\lambda}{d}\right) \bigg|_{m=1} = \sin^{-1}\left[\frac{(1)(\lambda)}{100\lambda}\right] = 0.010 \text{ rad.}$$

...5'

(b) Since $y_1 = D \tan \theta_1$ (see Fig. 35-10(a)), we obtain

$$y_1 = (900 \text{ mm}) \tan (0.010 \text{ rad}) = 9.0 \text{ mm.}$$

The separation is $\Delta y = y_1 - y_0 = y_1 - 0 = 9.0 \text{ mm.}$

...5'

Q7.**Solution:**

(a) The angular positions θ of the bright interference fringes are given by $d \sin \theta = m\lambda$, where d is the slit separation, λ is the wavelength, and m is an integer. The first diffraction minimum occurs at the angle θ_1 given by $a \sin \theta_1 = \lambda$, where a is the slit width. The diffraction peak extends from $-\theta_1$ to $+\theta_1$, so we should count the number of values of m for which $-\theta_1 < \theta < +\theta_1$, or, equivalently, the number of values of m for which $-\sin \theta_1 < \sin \theta < +\sin \theta_1$. This means $-1/a < m/d < 1/a$ or $-d/a < m < +d/a$. Now

$$d/a = (0.180 \times 10^{-3} \text{ m}) / (30.0 \times 10^{-6} \text{ m}) = 6.00,$$

so the values of m are $m = -5, -4, -3, -2, -1, 0, +1, +2, +3, +4$, and $+5$. There are 11 fringes.

...6'

(b) The intensity at the screen is given by

$$I = I_m (\cos^2 \beta) \left(\frac{\sin \alpha}{\alpha} \right)^2$$

where $\alpha = (\pi a / \lambda) \sin \theta$, $\beta = (\pi d / \lambda) \sin \theta$, and I_m is the intensity at the center of the pattern. For the third bright interference fringe, $d \sin \theta = 3\lambda$, so $\beta = 3\pi$ rad and $\cos^2 \beta = 1$. Similarly, $\alpha = 3\pi a / d = 3\pi / 6.00 = 0.500\pi$ rad and

$$\left(\frac{\sin \alpha}{\alpha} \right)^2 = \left(\frac{\sin 0.500\pi}{0.500\pi} \right)^2 = 0.406$$

The intensity ratio is $I/I_m = 0.406$.

...6'

Q8.

Solution:

a) In the armada's rest frame (called S_a), the velocity of the messenger is:

$$v' = \frac{v - v_a}{1 - vv_a / c^2} = \frac{0.95c - 0.85c}{1 - (0.95c)(0.85c) / c^2} = 0.52c$$

The length of the trip is proper length, and the time interval of the trip is

$$t' = \frac{L_0}{v'} = \frac{1.0 \text{ y}}{0.52c} = 1.925 \text{ y}.$$

...4'

(b) In the messenger's rest system (called S_m), the time interval of the trip is proper time:

$$\Delta t_0 = \frac{\Delta t}{\gamma} = 1.925 \text{ y} \sqrt{1 - 0.52^2} = 1.645 \text{ y}.$$

...4'